

MATERIAL SPECIFICATION

MS-2024-017 Rev B

Aluminum Alloy 6061-T6, Bar, Rod, and Extrusions

1. SCOPE

This specification covers aluminum alloy 6061-T6 in the form of bars, rods, wire, and extruded shapes for use in structural and mechanical applications. This material is suitable for aerospace applications requiring moderate to high strength, good corrosion resistance, and excellent machinability.

2. APPLICABLE DOCUMENTS

AMS 4027 — Aluminum Alloy Bars, Rods, and Wire (6061-T6)
ASTM B221 — Aluminum Alloy Extruded Bars and Shapes
AMS 2772 — Heat Treatment of Aluminum Alloy Raw Materials
ASTM E8 — Standard Test Methods for Tension Testing

3. CHEMICAL COMPOSITION (WEIGHT %)

Element	Min %	Max %
Silicon (Si)	0.40	0.80
Iron (Fe)	—	0.70
Copper (Cu)	0.15	0.40
Manganese (Mn)	—	0.15
Magnesium (Mg)	0.80	1.20
Chromium (Cr)	0.04	0.35
Zinc (Zn)	—	0.25
Titanium (Ti)	—	0.15
Aluminum (Al)	Remainder	—

4. MECHANICAL PROPERTIES

Property	Minimum Value	Test Method
Ultimate Tensile Strength	45.0 ksi (310 MPa)	ASTM E8
Yield Strength (0.2% offset)	40.0 ksi (276 MPa)	ASTM E8
Elongation (2 in. gauge)	10%	ASTM E8
Brinell Hardness	95 HB	ASTM E10

5. QUALITY ASSURANCE

Material shall be accompanied by a Certificate of Conformance (C of C) and a mill test report traceable to the original heat lot. Supplier shall maintain AS9100D quality management system certification.

Effective Date: August 15, 2024 | Approved by: J. Harrison, Quality Engineering